

Filobateer Ghaly

PhD Student

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Education

Ph.D: Physics	 2024 (current)  University of Calgary	Demonstrating the Trapping of Hydrogen in the HAICU Experiment
M.Sc.: Physics	 2022 - 2023  University of Calgary	Quantifying Ionospheric Electron Density Enhancements and their Impact on High Frequency Radio Wave Propagation
B.Sc.: Physics	 2017 - 2021  University of Calgary	

Research Contributions

HAICU Trap Commissioning – 2025–(present), Ph.D. Thesis

- Built various components of the magnetic trap.
- Tested the magnets with running cooling water and current up to 1200 Amps.
- Researching and developing strategies for integrating temperature sensors into the magnetic trap for monitoring and interlocking the magnets during operation.

HAICU Experiment Simulations – 2024–(present), Ph.D. Thesis

- Developed COMSOL Multiphysics simulations in conjunction with C++ code to model the magnetic field used in the HAICU experiment.
- Integrated the field maps into a C++ software package to simulate the trajectories of hydrogen in our apparatus.
- Used the simulations to time the trapping of hydrogen, optimizing for maximum number of atoms trapped.
- Extended the simulations to include magnetic deceleration of hydrogen.

Radio Communication Disruption Forecasting – 2023–2024, M.Sc. Thesis

- Performed an ordinal statistical analysis of over 4 years of radio and riometer data.
- By finding correlations between the two datasets, I established the riometer response as proxy for high-frequency radio communication disruption.
- Identified the core drivers for signal loss for a wide range of frequencies from 5 to 15 MHz.
- Findings currently under review for publication in the *Journal of Geophysical Research: Space Physics*.

Sizing Radio-disrupted Regions – 2022–2023, M.Sc. Thesis

- Wrote a ray tracing program to model the undisturbed propagation of radio waves.
- Used the modelled signal, together with the signal status observed by the receiver, to estimate the size of the region where radio waves are disrupted.
- Extended the methodology to a network of multiple radio transmitters and receivers across Canada.
- The technique was published in: *Journal of Geophysical Research: Space Physics*: Ghaly et al., 2024. All code, data analysis, visualizations and writing were done by me. Coauthors helped with editing the paper.

Collaboration and Impact

Natural Resources Canada (NRCan) Collaboration – 2020–2024 Research Assistant and M.Sc. Thesis

- Upon discovering NRCan's underutilized time-signal transmitters, I verified their detectability in our data.
- I then established that their signals were disrupted by space weather, and collaborated with NRCan to incorporate their network of radio transmitters into my analysis.
- The success of this collaboration led to the development of a space weather forecasting/alerting system for modeling where and when radio communications are disrupted.

Software and Technical Development

MITACS Globalink Award: CERN Control System Revamp – Summer 2025

- I was awarded a MITACS Globalink Research Award to spend the summer of 2025 at CERN, revamping the control system for the microwaves in the ALPHA experiment.
- Upgrading from a National Instruments PXI crate to a compact-RIO, I developed entirely new control software in LabVIEW.
- The new software uses the Remote Action Engine framework, allowing for remote control and monitoring of the microwave system.

Real-time Data Pipeline and Automatic Baselineing – 2021–2023, Research Assistant and M.Sc. Thesis

- Adapted the Perseus, Software-Defined Radio (SDR), making it usable for research.
- Wrote software for data acquisition in Python and Java and tested the instrument's ability to function as a receiver.
- Developed an online data browser for the prototype instruments deployed to generate curated, raw (K0-level) data for the public. Back-processed and published K0 data for 2022 and 2023.
- Wrote instrument-side code to generate curated K0 data on-site for all instruments deployed. Data are now being published in real-time.
- Developed a Graphical User Interface (GUI) for semi-automatic, instrument-specific baselineing.
- Developed a methodology for real-time, automatic baselineing to generate operational (K2-level) data.
- Designed a web application with a Postgre database that browses through different specification parameters for flagging space weather events.

Scholarships and Awards

Award Name	Year	Amount (CAD)
Alberta Graduate Excellence Scholarships	2025	15,000
Mitacs Globalink Research Award	2025	6,000
Indigenous and Black Momentum Scholarship in Science	2024	25,000
International Graduate Tuition Award	2023	3,000
Institute for Space Research Graduate Scholarship in Space Physics	2023	1,100
International Graduate Tuition Award	2022	3,000
University of Calgary Undergraduate Competition Award	2021	3,600
President's Admission Scholarship	2015	4,000
University of Calgary Engineering Alumni Scholarship	2015	5,900
University of Calgary International Entrance Award	2015	10,000

Publications

MSc Thesis

- **Sept. 2024** F. Ghaly. "Quantifying Ionospheric Electron Density Enhancements and their Impact on High Frequency Radio Wave Propagation". en. URL: <https://hdl.handle.net/1880/119764>.

Articles

- **2024** F. Ghaly et al. "Use of Terrestrial High Frequency Signals in Riometer Data to Explore the Size of D-Region Electron Density Enhancements". en. In: *Journal of Geophysical Research: Space Physics* 129.5. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1029/2023JA032375>, e2023JA032375. ISSN: 2169-9402. DOI: 10.1029/2023JA032375. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1029/2023JA032375>.

Reports

- **Mar. 2022** F. Ghaly and E. Spanswick. *Identification and Investigation of High Frequency Radio Wave Transmissions During Quiet and Disturbed Periods*. Tech. rep. Natural Resources Canada.

Presentations

- **Feb. 2025** F. Ghaly. "Towards Demonstrating Trapping of Hydrogen in HAICU". *Winter Nuclear and Particle Physics Conference (WNPPC)*. Banff, Alberta.
- **Dec. 2023** F. Ghaly et al. "A Riometer Proxy for HF Terrestrial Link Status". *American Geophysical Union (AGU) 2023 Fall Meeting – (Poster)*. San Francisco, California.
- **Feb. 2023** F. Ghaly et al. "Using Multi-frequency Riometers for Monitoring HF Signal Propagation And Modelling D-region Absorption". *Division of Atmospheric and Space Physics Conferences (DASP)*. Canadian Space Agency (CSA). John H. Chapman Space Centre, Longueuil, Quebec.
- **Dec. 2022** F. Ghaly et al. "Use of Terrestrial HF Signals in Riometer Data to Investigate the Size of the Absorbing Region". *American Geophysical Union (AGU) 2022 Fall Meeting – (Poster)*. Chicago, Illinois.
- **Feb. 2021** F. Ghaly, E. Spanswick, and R. Gillies. "Investigating Riometer Capacity for Terrestrial HF Signal Monitoring". *Division of Atmospheric and Space Physics Conferences (DASP)*. Online via Zoom.

Teaching Experience

Teaching Assistant (TA): Department of Physics and Astronomy  University of Calgary

Admin TA	PHYS 259: Electricity and Magnetism		Winter, 2025
Lab TA	PHYS 211: Mechanics		Fall, 2024
Lead TA	PHYS 355: ELectromagnetic Theory		Winter, 2024
Lab TA	PHYS 397: Applied Physics Labratory		Fall, 2023
Lab TA	PHYS 229: Modern Physics		Winter, 2021
Lead TA	PHYS 259: Electricity and Magnetism		Summer, 2020
Lab TA	PHYS 259: Electricity and Magnetism		Winter, 2020

Learning Assistant (LA): Department of Computer Science  University of Calgary

Tutorial LA	DATA 202: Thinking with Data		Winter, 2021
Tutorial LA	CPSC 501: Advanced Programming Techniques		Fall, 2020

*As lead TA for the remote restructuring of the PHYS259 laboratory curriculum, right after the onset of the COVID pandemic (Summer 2020), I transformed physical lab activities for remote delivery, expanded a near-empty online question bank by over 100 items with cheat-resistant quiz variants, and—on my own initiative—produced weekly tutorial videos solving four to seven problems in detail. These videos provided continuity of support to students who had lost access to in-person instruction entirely.

Skills

Software/ Programming:

• Python

• Java

• JavaScript

• C++

• MATLAB

• GNU Radio Companion

• PostgreSQL

• SQLite

• Django

• CSS

• HTML

Hardware:

• Software Defined Radios (SDR)

• Antennae

• Microcontrollers

• Machine Shop Trained

Modelling/ Simulations:

• AutoCAD

• Inventor

• COM-SOL Multiphysics

Languages

Arabic	C2*	
English	C2*	